

Megha Narendra Simha

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Professional Summary

Software Engineer with 4+ years of experience building AI-powered security tooling and scalable production systems. Hands-on experience verifying AI agent behavior with automated guardrails (prompt injection, hallucination, output validation) before outputs are trusted, plus building cryptographic threat-detection pipelines and automating security workflows in CI/CD. Skilled across cloud platforms (AWS, Azure, GCP), object-oriented design, and cross-functional collaboration, with a strong foundation in AI/ML (PyTorch, TensorFlow) applied to real-world security problems.

Professional Experience

AI Security Intern, iLink Digital Apr 2026 – Present

- Built an AI-powered PQC security assessment pipeline verifying cryptographic risk and AI agent output before deployment (see Projects), reducing solution design time by **40%**.
- Evaluated **15+ PQC and CBOM concepts**, detecting quantum-vulnerable algorithms (**Shor's, Grover's, HNDL**) and threats requiring remediation.
- Assessed VPN quantum risks and hybrid **TLS 1.3 + PQC** models, strengthening endpoint and infrastructure threat detection expertise.

Student Worker Teaching Assistant, City University of Seattle Jan 2025 – Mar 2026

- Led support for AI, Database Systems, Cloud Computing, and Secure Systems courses for **200+ students** and 8 professors.
- Organized and executed a hackathon with **50+ participants**, mentoring teams on AI agent and scalable system solutions.

Software Developer Intern, Costco Wholesale Jun 2024 – Aug 2024

- Improved .NET Core microservices by optimizing async workflows and SQL queries, reducing API latency by **30%** while maintaining secure data handling practices.
- Built secure REST APIs using OAuth2, JWT, and RBAC, enforcing authentication, authorization, and access control for **8,000+ requests/day**.
- Presented system architecture to **50+ stakeholders** in a fast-paced, cross-functional environment.

Application Development Analyst, Accenture Feb 2022 – Feb 2023

- Developed backend and distributed data systems for **10,000+ users**, applying object-oriented design and secure coding practices.
- Designed QR-based workflows and optimized frontend performance, reducing load time by **40%**.
- Owned operational health and deployment efficiency using **Azure DevOps CI/CD** and **Cosmos DB**, including monitoring and incident resolution.

Software Developer, NTT Data Apr 2019 – Mar 2021

- Led RxChange feature development in a multi-tiered distributed system and optimized SQL queries, reducing latency by **30%**.
- Delivered documentation, code reviews, and KT sessions for **50+ engineers**, and demoed product capabilities to **100+ stakeholders**.

AI Projects & Research

PQC Security Assessment Workspace - AI Security Copilot with Guardrails

- Built an AI agent (Mistral LLM) that automatically verifies cryptographic risk and generates security reports, gated by **3 production guardrails** – prompt injection protection, hallucination detection, and output validation – **before** results reach engineers.
- Integrated **CBOM parsing**, SonarQube vulnerability data, and a NIST-grounded **RAG** pipeline (LangChain, ChromaDB) to detect quantum-vulnerable algorithms and auto-generate prioritized remediation plans; automated end-to-end in a **GitHub Actions CI/CD** security gate that blocks merges below a security threshold.

SmartCart AI - A Decision-Centric Agentic System for Inventory Waste Reduction and Pricing Optimization

- Designed a multi-agent system with secure RAG pipelines (PGVector, PyTorch, **OWASP-aligned**), reducing perishable waste by **92%** and improving prediction accuracy by **85%**; authored an ISCAP research paper.

Supply Soul - Pharmacy Inventory Management

- Built a multi-agent inventory forecasting system (FastMCP, PGVector) with a voice AI assistant and automated workflows, improving stock availability by **30%** and reducing stockouts by **50%**.

Education

M.S. Computer Science, City University of Seattle (**Dean's List, Top 10%**) Apr 2024 – Mar 2026

B.E. Computer Science, JSSATE Bengaluru Jun 2014 – Jun 2018

Technical Skills

AI Agent Verification & Security: AI Guardrails (Prompt Injection, Hallucination, Output Validation), Threat Detection, OWASP-Aligned Secure Design, PQC, CBOM

AI, LLM & Agent Systems: Multi-Agent Systems, MCP Servers (FastMCP), LangChain, RAG, Prompt Engineering, Mistral/LLM Integration

Languages & OOD: Python, C#, Java, JavaScript, TypeScript, Object-Oriented Design

Cloud & CI/CD: AWS, Azure, GCP, GitHub Actions, CI/CD Automation, Microservices, Distributed Systems

Security & Databases: OAuth2, JWT, RBAC, ChromaDB, PostgreSQL, SQL Server, MongoDB, Redis